Scalable manufacturing architectures for heterogeneously integrated, injection-locked PCSEL chiplet arrays

Experimental scaling laws and wafer-compatible pathways toward megawatt-class coherent apertures

Proposed PhD at the Australian National University, with Kyoto University as the preferred PCSEL device partner

THESIS IN ONE SENTENCE

Use a reconfigurable array of independently controllable PCSEL chiplets to measure the locking, thermal, phase and fault laws that govern coherent-domain scaling, then translate - without overclaiming - those laws into requirements for future low-cost semiconductor manufacture.

WHY THIS RESEARCH IS TIMELY

PCSEL device physics is advancing rapidly. A 3 mm device has exceeded 50 W continuous-wave single-mode output; theory describes routes from 100 W toward 1 kW; a 10 mm resonator has produced 500 W in pulsed operation; and injection locking has been demonstrated using a seed several orders weaker than the slave output.[1-4] A four-device module has produced 200 W continuous-wave output, but not demonstrated mutual phase coherence.[5] Small two- to four-element monolithic arrays have demonstrated controllable coherence and coherent power-scaling effects at much lower power.[6,7]

This establishes credible device mechanisms, but not the scalable coherent machine. The current bounded public record does not demonstrate a common-seeded independent-chiplet array, a measured channel-count law, phase stitching between PCSEL domains, full-field fault containment or a low-cost production architecture.

RESEARCH QUESTIONS

Primary experimental question

Under what interface, injection-locking, thermal and control architectures can independently controllable PCSEL chiplets form coherent domains and hierarchically stitched arrays whose phase stability, useful-order performance and fault response remain predictable as channel count N grows?

Secondary manufacturing question

What constraints do those measured laws impose on scalable, low-cost monolithic-wafer, reticle-domain, known-good-die and hybrid manufacture?

WORKING HYPOTHESIS

A reconfigurable chiplet array will exhibit a finite, measurable coherent-domain frontier, N^{*}_{domain}, set by device detuning, thermal covariance, seed ratio, trim authority and control residual. Below it, local coherence should remain predictable; qualified domains may then be composable through a slower phase-stitching loop.

The chiplet array is the experimental instrument, not a commitment to die-by-die production. A negative or unexpectedly small frontier remains a valid doctoral result if it is repeatable, predictive and mechanistically explained.

FOUR WORK PACKAGES

WP1 - Chiplet interface, locking and thermal population law. Define the observable interface. Measure locking, phase, acquisition, recovery and cross-channel thermal response against detuning, temperature, seed ratio and trim; test prediction on held-out devices or operating points.

WP2 - Physical coherent-domain scaling. Add and reconfigure chiplets while measuring lock probability, phase covariance, control margin and useful-order performance. Infer N^{*}_{domain} over the tested envelope.

WP3 - Phase stitching, complete fields and faults. Stabilise at least two domains, then close a slower outer piston loop. Measure complete fields under device, seed-branch, thermal, phase-bias and domain faults; identify which remain local.

WP4 - SCALA and manufacturing translation. Calibrate and validate a reproducible model, then compare monolithic, reticle-domain, known-good-die and hybrid pathways. Output necessary conditions and conditional yield/cost sensitivities - not a qualified process or demonstrated factory economics.

A FEASIBLE DOCTORAL SCALE

EVIDENCE LEVEL	PHYSICAL SCOPE	CLAIM SUPPORTED
Minimum	2-8 independently controllable real PCSEL chiplets	PCSEL-specific interface, population, first-domain and two-domain stitching evidence over the tested envelope
Strong	16-64 real or mixed real/emulated channels, with emulation calibrated by physical measurements	Stronger channel/domain scaling, complete-field prediction and fault-containment evidence
Physical stretch	Approximately 100 observable real channels	Extension of the measured physical range only
Controls stretch	At least 1,000 hardware-in-the-loop channels	Controller timing, telemetry and fault logic only - no unmeasured optical, thermal or manufacturing claim

Written access to at least 2-8 independently controllable real PCSELs, sufficient device information, a safe operating location and publishable measurement rights is a pre-start gate. Surrogates can commission the apparatus, but cannot establish the PCSEL-specific result.

The doctorate does **not** require a 1 kW continuous-wave PCSEL, a 100 kW tile, a megawatt wafer or aperture, production cooling and electrical buses, lifetime qualification, factory economics, atmospheric propagation or safety certification. Cross-supplier interchangeability is optional evidence, not a premise.

PROPOSED ANU-KYOTO STRUCTURE

ANU as doctoral home. The proposed ANU contribution is heterogeneous integration, reconfigurable packaging, coherent control, thermal identification, complete-field metrology, SCALA and manufacturing translation. Public profiles indicate relevant activity in silicon-nitride hybridisation and high-power lasers, adaptive-optics instrumentation and III-V light sources on silicon.[8] This indicates potential fit, not commitment or facility availability.

Kyoto as preferred device partner. Kyoto's PCSEL Research Center is the natural setting for device-specific collaboration.[9] The proposed contribution is access to, or joint definition of, controllable PCSEL research articles, operating limits and interpretation. No device supply or agreement is assumed.

This keeps the doctoral contribution above the die while grounding it in real measurements. IP, sample/data rights, publication and cross-border terms would precede restricted exchange.

WHAT I AM SEEKING FROM AN INITIAL MEETING

1. Test the novelty, falsifiability and doctoral scope.
2. Identify an ANU panel spanning heterogeneous photonics, coherent control/metrology and semiconductor emitters.
3. Determine whether ANU can host the minimum experiment and what facilities and approvals it needs.
4. Define the smallest useful Kyoto collaboration: 2-8 controllable PCSELs, interface information, operating constraints and interpretation.
5. Scope the first experiment: characterise two or more PCSELs, form and stitch two domains, insert one perturbation, and measure survivor state and the complete field.

SELECTED PUBLIC EVIDENCE

1. T. Inoue *et al.*, *Nature Communications* 13, 3262 (2022), doi:10.1038/s41467-022-30910-7. Theoretical 100 W-1 kW design framework, not a demonstrated 1 kW device.
2. M. Yoshida *et al.*, *Nature* 618, 727-732 (2023), doi:10.1038/s41586-023-06059-8.
3. M. Yoshida *et al.*, CLEO 2025, SS127_6, doi:10.1364/CLEO_SI.2025.SS127_6. The 500 W result is pulsed.
4. T. Inoue *et al.*, CLEO 2025, SS127_5, doi:10.1364/CLEO_SI.2025.SS127_5.
5. Y. Fukada *et al.*, *IEEE JSTQE* 32(6), 4900307 (2026), doi:10.1109/JSTQE.2026.3705514. The 2 × 2 module does not establish mutual coherence.
6. R. J. E. Taylor *et al.*, *Scientific Reports* 5, 13203 (2015), doi:10.1038/srep13203.
7. B. C. King *et al.*, *AIP Advances* 11, 015017 (2021), doi:10.1063/5.0031158.
8. ANU public profiles: Steve Madden, Céline d'Orgeville and Hoe Tan.
9. Kyoto University PCSEL Research Center.